



University of Colombo, Sri Lanka

University of Colombo School of Computing

BIT

**DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**

Academic Year 2025 — 1st Year Examination — Semester 1

IT1506 — Fundamentals of Mathematics

Multiple Choice Question Paper
(1 Hour)

Important Instructions

- The duration of the paper is **1 Hour**.
- The medium of instructions and questions is English.
- This paper has **20 questions** on **4 pages**. Answer **all** questions.
- All questions are of the **MCQ** (Multiple Choice Questions) type.
- Each question will have **5 (five)** choices with **ONLY ONE** correct answer.
- This paper consists of 100 marks and all the questions will carry equal marks.
- Answers should be marked on the **special answer sheet** provided.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigilator immediately.
- Mark the correct choices on the question paper first and then transfer them to the given answer sheet which will be machine marked. **Please completely read and follow the instructions given on the other side of the answer sheet before you shade your correct choices.**
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are not allowed.
- **All Rights Reserved.** This question paper can **NOT** be used without proper permission from the University of Colombo School of Computing.

1) The solution to the equation $16^{(x-3)} \times 2^{(5x+1)} = 8^6$ is

(a) $9/26$
(d) $4/11$

(b) 26
(e) $29/9$

(c) 9

2) What is the decimal representation of $(\frac{3}{16} + \frac{3}{32} + \frac{11}{8})$?

(a) 1.656
(d) 2.557

(b) 1.687
(e) 0.165

(c) 2.555

3) Convert $1\frac{2}{7}$ into a percentage:

(a) 128.57 %
(d) 1.2857%

(b) 28.57 %
(e) 2.857 %

(c) 18.57%

4) What is the Least Common Multiple (LCM) of 16, 36 and 12?

(a) 108
(d) 144

(b) 96
(e) 192

(c) 72

5) For two natural numbers a and b, $gcd(a, b) \times lcm(a, b)$ is equal to:

(a) $a + b$
(d) $a \times b$

(b) a/b
(e) $a + a/b$

(c) $a^2 + b^2$

6) Simplified representation of $\frac{4x^2-1}{x+1} + \frac{x^2+x-3}{x-1}$ is

(a) $(5x^3 - 2x^2 - 3x - 2)/(x^2 - 1)$
(c) $(4x^2 + 9x + 1)/(x^2 - 1)$
(e) $(2x^2 + 2x + 1)/(x + 1)$

(b) $(4x^2 + 4x + 1)/(x^2 - 1)$
(d) $5x^3 - 2x^2 - 3x - 2/(x^2 + 1)$

7) Oil from a hemispherical tank of radius 14 cm is poured into a cylindrical vessel of radius 7 cm. Find the height of oil in the cylinder.

(a) $28/9$ cm
(d) $21/32$ cm

(b) $370/9$ cm
(e) $112/3$ cm

(c) $720/9$ cm

8) Factorize $x^3 - 2x^2 - 15x + 30$

(a) $(x + 3)(x^2 - 5x + 10)$
(d) $(x - 3)(x^2 + x - 10)$

(b) $(2x + 2)(x - 3)$
(e) $(x + 6)(x^2 - 5)$

(c) $(x - 2)(x^2 - 15)$

9) What is the solution of $3x^2 - 5x - 7 = 0$?

- (a) $x = 3, -1$ (b) $x = (6 \pm \sqrt{109})/5$ (c) $x = \pm\sqrt{109}$
 (d) $x = (5 \pm \sqrt{109})/6$ (e) $x = -13, 11$

10) $Q = 5(\frac{N}{4} + M + K)^{1/3}$. Which of the following correctly expresses N as the subject of the formula?

- (a) $N = 3(\frac{Q^4}{125} - M - K)$ (b) $N = 4(\frac{Q^3}{125} + M - K)$ (c) $N = 3(\frac{Q^3}{125} - 4(M - K))$
 (d) $N = 4(\frac{Q^3}{125} - M - K)$ (e) $N = 3(\frac{Q^3}{125} - M - K)$

11) Consider the quadratic equation $x^2 - 2x - 3 = y$ and the line $y = x + 1$. What is/are the point(s) of intersection of the curve and the line.

- (a) (2,3) & (1,3) (b) (1,1) & (3,5) (c) (0,1) & (-1, -3)
 (d) (4,5) & (-1,0) (e) (2,3) & (-1, -3)

12) Kamal purchased two lands, L1 and L2, for Rs. 1 million and Rs. 2 million, respectively. He later sold L1 for Rs. 1.2 million and L2 for Rs. 2.75 million. What is the ratio of the total profit made from both land sales to the profit made from land L1?

- (a) 5:1 (b) 4:19 (c) 9:4
 (d) 19:4 (e) 19:2

13) Which of the following functions has a domain $x > 1$, a vertical asymptote at $x = 1$, and passes through the point (2,0)?

- (a) $y = \ln x$ (b) $y = e^{x-1}$ (c) $y = \ln(1 - x)$
 (d) $y = \ln(x - 1)$ (e) $y = \log(x + 1)$

14) What is the domain of $f(x) = (\sqrt{x^2 - 4})/x$?

- (a) $x \geq 2 \cup x < 2$ (b) $x \leq -2 \cup x \geq 2 \cup \{0\}$ (c) $\mathbb{R}$
 (d) $x < 4$ (e) $x \leq -2 \cup x \geq 2$

15) Which of the following statements correctly compares the functions, $y = e^x$ and $y = \ln x$

- (a) Both functions have a vertical asymptote at $x = 0$
 (b) $y = \ln x$ grows faster than $y = e^x$ for large x
 (c) $y = \ln x$ is the inverse of $y = e^x$
 (d) Both functions are defined for all real x
 (e) Both functions pass through the point (0,1)

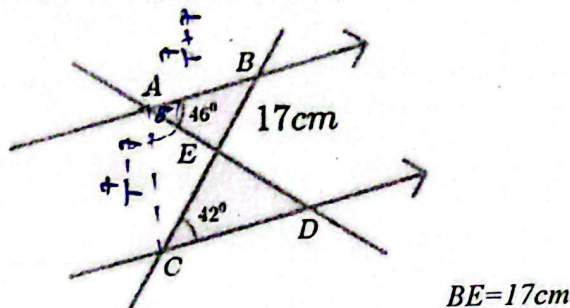
- 16) Using the BODMAS rule, determine the result of $6 + 3 \times (10 - 4) \div 2 + 1$?

(a) 13 (b) 12 (c) 16
(d) 14 (e) 15

- 17) The sides of a triangle are x , $x + 4$, and $x + 8$. If the triangle is right-angled, find the value of x .

(a) 4 (b) 8 (c) 12
(d) 6 (e) 10

Consider the following diagram and answer Question 18.



- 18) What is the length of AB in cm?

(a) $\{17/\sin(46)\} \times \sin(92)$ (b) $\{17/\sin(46)\} \times \sin(8)$ (c) $\{17/\sin(92)\} \times \sin(46)$
(d) $\{17/\sin(138)\} \times \sin(46)$ (e) $\{17/\sin(46)\} \times \sin(8)$

- 19) In a triangle ABC , sides $AB = (t + \frac{1}{t})$, $AC = (t - \frac{1}{t})$, and the included angle $\hat{A} = 60^\circ$. Find BC .

(a) $t^2 + (\frac{3}{t^2})$ (b) $\{(t^2 - 1)^2\}/t^2$ (c) $\{(t^2 + 1)^2\}/4t^2$
(d) $\{(t^2 - t)^2\}/4t^2$ (e) $t^2 - (\frac{3}{t^2})$

- 20) If $\cos x = t$ then $(\sin x / \cos x) + \sec x$ is equal to:

(a) $(1 + \sqrt{1 - t^2})/t$ (b) $1 + \sqrt{1 - t^2}/t^2$ (c) $(t^2 - 1)^2/2t^2$
(d) $(t^2 - 1)^2/t^2$ (e) $(t^2 + 1)^2/2t^2$
